# ETANERCEPT

Screen for TB & hepatitis B

Weekly SC injection

**CHEMICAL NAME** Recombinant fusion protein: extracellular domain of human TNF receptor p75 linked to Fc portion of human IgG1

**CLASS** Biologic — TNF-α inhibitor (soluble receptor fusion protein)

**TRADE NAMES** Enbrel · Benepali · Erelzi (biosimilars)

## THERAPEUTIC USES / INDICATIONS

- Rheumatoid arthritis, psoriatic arthritis, ankylosing spondylitis.
- Juvenile idiopathic arthritis (from 2 yr).
- Plaque psoriasis (adults and children ≥6 yr).
- Not effective for inflammatory bowel disease.

## ADMINISTRATION

- **SC** injection weekly (or twice weekly) into thigh, abdomen or upper arm; rotate sites.
- Store in fridge; warm to room temperature before use.

## DOSAGES

- **RA/PsA/AS:** 50 mg once weekly (or 25 mg twice weekly).
- **Psoriasis:** 50 mg twice weekly for 12 weeks, then 50 mg weekly.
- **JIA:** 0.8 mg/kg weekly (max 50 mg).

## PHARMACOKINETICS

- **Onset:** 1–2 weeks; peak benefit 3 months.
- Bioavailability ~76% SC.
- **t½:** ~4 days.

## NURSING MANAGEMENT

- Pre-treatment screening (TB, HBV, HCV, HIV); vaccinations.
- Teach weekly self-injection; sharps disposal.
- Monitor infection, FBC, disease activity.

## PHARMACODYNAMICS

- Acts as a decoy receptor; binds **TNF-α (and lymphotoxin-α)** → prevents interaction with cell-surface receptors → ↓ inflammation.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Injection-site reactions (common).
- Upper respiratory infections.
- Headache, rash.

### ADVERSE EFFECTS (SERIOUS)

- Serious infections; TB reactivation; hepatitis B reactivation.
- Lymphoma; skin cancers.
- Demyelination; heart failure worsening.
- Pancytopenia, aplastic anaemia (rare); lupus-like syndrome.

## CONTRAINDICATIONS & CAUTIONS

- Sepsis or risk of sepsis; active infections incl. TB.
- Live vaccines.

### KEY INTERACTIONS

- **Anakinra, abatacept:** ↑ serious infections — avoid.
- **Live vaccines:** contraindicated.
- **Cyclophosphamide:** ↑ malignancy (Wegener's study).

## PATIENT EDUCATION

- Inject on the same day each week.
- Report fever, persistent cough, bruising, bleeding or pale skin.
- Avoid live vaccines; inform other healthcare providers.

**KEY POINTS**

- TNF receptor–Fc fusion protein.
- Screen TB/HBV; infection risk.
- 50 mg SC weekly.
- Not used for IBD.